

MicrobeNetNet - Connecting Datasets/Nodes/Networks Across Disciplinary Siloes

as reflected in data connections

Jorrit Poelen (UC Santa Barbara Cheadle Center, Ronin Institute)

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Guiding Questions

Hey, are we unsiloed yet?

Do *you* have examples of unsilo-ing?

How do *you* think we should unsilo?

Ben's Challenge to Jorrit

[...] New Datasets/Nodes/Networks that might be connected (Jorrit)

- 1. What data are there (plain language summaries)?*
- 2. How can people access what are there?*
- 3. Case study, Review example of FRED and MaarjAM networks - how you found the data, used them. Generalizable for other efforts? [...] ¹*

¹Sikes, B. 2026-09-08. Pers. Comm.

Are we unsiloed yet? Revisiting MN^N Theme and Grand Challenge.

[...] MICROBENet²[1] Theme and Grand Challenge: Plants and microorganisms regulate a diverse array of processes that modulate the Earth system. Yet, we know remarkably little about how variable plant-microbial interactions are across the globe, how microbial connections with plants respond to global change, and if changes in plant-microbe interactions will lead to predictable changes in plant, ecosystem, and macro-scale ecology. While microbial ecologists and plant ecologists both explore questions about geography, interactions, and function, this work has often occurred in plant or microbial silos. **Focusing on either plants or microbes resulted in the development of distinct disciplinary vocabularies and meanings, even when using similar tools and asking questions across similar scales.** [...]

²Sikes, B.A., Zanne E.A., Classen, A.T, Kivlin S.N. (2024) MICROBENet-Net: Multi-Institute Collaborative Research on BELOWground plant-microbial interactions Network of Networks. 2024/2028 NSF AccelNet Award 2412561 <https://microbenetwork.net/>
<https://globalbioticinteractions.org/microbenetnet>

Are we unsiloed yet? MN^N 2025 Colloquium in Knoxville, TN.



3.

³Sikes, B., Classen, A., Kivlin, S., Zanne, A. & Holly Andres. (2025, May 29). 2025 MicrobeNet[^]Net Colloquium. Zenodo. 2025 MicrobeNet[^]Net Colloquium, Knoxville, TN.
<https://doi.org/10.5281/zenodo.15547951>

Are we unsiloed yet? MN^N 2025 Colloquium in Knoxville, TN. FRED and MaarjAM meet.



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⁴Sikes, B., Classen, A., Kivlin, S., Zanne, A. & Holly Andres. (2025, May 29). 2025 MicrobeNet[^]Net Colloquium. Zenodo. 2025 MicrobeNet[^]Net Colloquium, Knoxville, TN.
<https://doi.org/10.5281/zenodo.15517051>

Are we unsiloed yet? FRED and MaarjAM

- ▶ FRED/MaarjAM met in person and data (csv, xlsx->csv) was shared in person
- ▶ FRED/MaarjAM data openly published
 - ▶ <https://github.com/globalbioticinteractions/fred>
 - ▶ <https://github.com/globalbioticinteractions/maarjam>
- ▶ GloBI integration tools used to connect FRED/MaarjAM via plant taxonomic names (World of Flora Online)
- ▶ Early applications include GloBI Data Reviews and Plant Overlap Analysis.
 - ▶ Elton, Nomer, & Preston. (2026). Versioned Archive and Review of Biotic Interactions and Taxon Names Found within globalbioticinteractions/fred hash://md5/6db551793883f575d828d8a600e1af72. Zenodo.
<https://doi.org/10.5281/zenodo.20547776>
 - ▶ Elton, Nomer, & Preston. (2026). Versioned Archive and Review of Biotic Interactions and Taxon Names Found within globalbioticinteractions/maarjam hash://md5/4578fcd83fceba1f11c269d8aaf1118a. Zenodo.
<https://doi.org/10.5281/zenodo.20548434>
 - ▶ <https://github.com/jhpoelen/fungal-plant-host-overlap/tree/0faeeae123b5d8f53576feb1d607ffb5d2bd3ac>

Are we unsiloed yet? A “Plain Language” (NSF) Update

*Since the inaugural MicrobeNetNet ([1], MN^N) colloquium in Knoxville, TN 2025-03-20/2025-03-24 [2], Jorrit Poelen has collaborated with many MN^N participants and nodes to use existing methods [3,4] to review and integrate existing datasets, databases, and collections describing plant-microbe interactions into the openly accessible and citable MicrobeNetNet Data Review Corpus [5,6]. **The 2026-08-14 version of the MN^N Review Corpus is a global dataset containing over 7 million claims sourced from MycoPortal (a data portal of a consortium of Mycology Collections), USDA United States National Fungus Collections Fungus-Host Dataset, Microbial Atlas of Predicted Species (MAPS), AusTrait, Fine Root Ecology Database (FRED), MaarjAM, FungalTraits, FunFun, Global(AM)Fungi, International Collection of Microorganisms from Plants (ICMP), MycoDB, USDA ARS Culture Collection (NRRL), and Unite and more. [...]** We expect to continue to integrate existing datasets [...], but also improve the data infrastructures [7] that help keep this openly data accessible.*⁵

⁵Excerpt from Jorrit's update submitted to Ben for NSF report on 2026-08-14

Are we unsiloed yet? MN^N Dataset Review Corpus

Published August 14, 2026 | Version v3

Dataset Open

MicrobeNetNet Dataset Review Corpus and Associated Data
Products hash://md5/5981a37a16c25204dc18d9188f185b19

hash://
sha256/7448991edd5f79db02519a5b3d4691c50a60c02f45d40f08
7402a7a492da3a3e

Poelen, Jorrit H. (Editor)

This repository provides workflow for building an integrated review corpus of datasets reviewed by Global Biotic Interactions (GloBI, Poelen et al. 2014).

The examples below are applied to the collection of datasets mentioned in various meetings facilitated by MicrobeNetNet (Sikes et al. 2024) and includes about 7M interactions claims extracted from over 60 datasets.

These examples show that automated data review is possible at scale across heterogeneous datasets of known origin.

filename	description
data.zip	Preston archive containing datasets under review
biblio.bib	reference list of datasets under review (bibtex)
...	tab-separated value formatted text file containing interaction claims extracted by Elton, a GloBI

Manage

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New version

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Versions

Version v3

Aug 14, 2026

10.5281/zenodo.21940611

Version v2

Feb 26, 2026

10.5281/zenodo.18794828

Version v1

Feb 26, 2026

10.5281/zenodo.18793975

⁶Poelen, J. H. (2026). MicrobeNetNet Dataset Review Corpus and Associated Data Products
hash://md5/5981a37a16c25204dc18d9188f185b19

Are we unsiloed yet? MN^N Dataset Review Corpus

(You or Your Colleagues)

-[:curate]->

(Original Data)

-[:usedIn]->

(MNN Data Review)

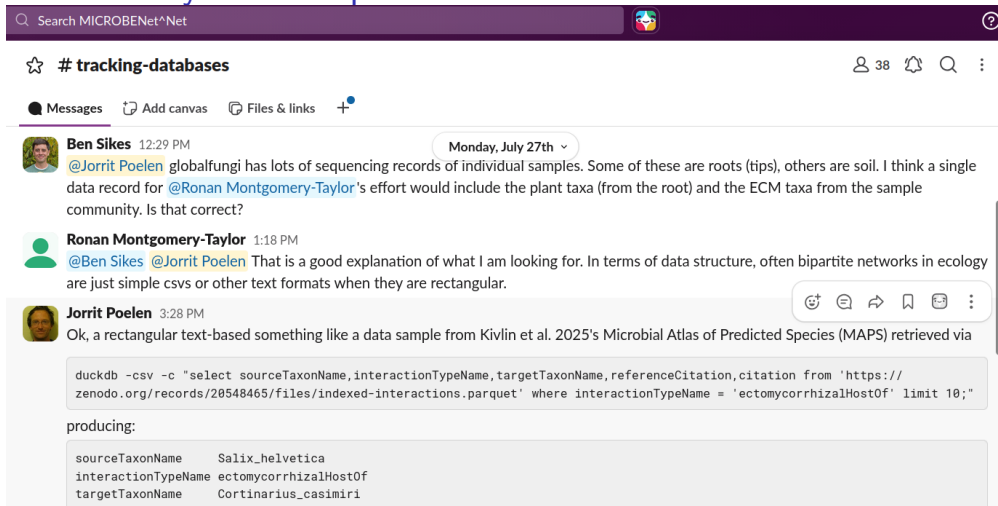
-[:partOf]->

(MNN Data Review Corpus)

-[:usedBy]->

(... everyone?)

Are we unsiloed yet? Example use.



Search MICROBENet^Net

☆ # tracking-databases

Messages Add canvas Files & links +

Monday, July 27th ▾

Ben Sikes 12:29 PM
@Jorrit Poelen globalfungi has lots of sequencing records of individual samples. Some of these are roots (tips), others are soil. I think a single data record for @Ronan Montgomery-Taylor's effort would include the plant taxa (from the root) and the ECM taxa from the sample community. Is that correct?

Ronan Montgomery-Taylor 1:18 PM
@Ben Sikes @Jorrit Poelen That is a good explanation of what I am looking for. In terms of data structure, often bipartite networks in ecology are just simple csvs or other text formats when they are rectangular.

Jorrit Poelen 3:28 PM
Ok, a rectangular text-based something like a data sample from Kivlin et al. 2025's Microbial Atlas of Predicted Species (MAPS) retrieved via

```
duckdb -csv -c "select sourceTaxonName,interactionTypeName,targetTaxonName,referenceCitation,citation from 'https://zenodo.org/records/20548465/files/indexed-interactions.parquet' where interactionTypeName = 'ectomycorrhizalHostOf' limit 10;"
```

producing:

sourceTaxonName	Salix_helvetica
interactionTypeName	ectomycorrhizalHostOf
targetTaxonName	Cortinarius_casimiri

Figure 1: MN^N Slack Exchange in #tracking-databases channel on 2026-07-27 as accessed at <https://app.slack.com/client/T068C6GQ1MW/C0AH89QFH9T> on 2026-09-11

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References

- [1] Sikes, B.A., Zanne E.A., Classen, A.T, Kivlin S.N. (2024) MICROBENet-Net: Multi-Institute Collaborative Research on BELOWground plant-microbial interactions Network of Networks. 2024/2028 NSF AccelNet Award 2412561
<https://microbenetwork.net/> <https://globalbioticinteractions.org/microbenetnet>
- [2] Sikes, B., Classen, A., Kivlin, S., Zanne, A.& Holly Andres. (2025, May 29). 2025 MicrobeNet^Net Colloquium. Zenodo. 2025 MicrobeNet^Net Colloquium, Knoxville, TN. <https://doi.org/10.5281/zenodo.15547951>
- [3] Jorrit H. Poelen, James D. Simons and Chris J. Mungall. (2014). Global Biotic Interactions: An open infrastructure to share and analyze species-interaction datasets. Ecological Informatics. <https://doi.org/10.1016/j.ecoinf.2014.08.005>.
- [4] Elliott M.J., Poelen, J.H. & Fortes, J.A.B. (2023) Signing data citations enables data verification and citation persistence. Sci Data. <https://doi.org/10.1038/s41597-023-02230-y>
hash://sha256/f849c870565f608899f183ca261365dce9c9f1c5441b1c779e0db49df9c2a19d
- [5] Poelen, J. H. (2026, Feb). MicrobeNetNet Dataset Review